

CER-AI PREOPERATIVE ECTASIA RISK ASSESSMENT

Corneal Ectasia Risk Assessment Intelligence

The final surgical decision and all associated responsibility and liability rest with the surgeon. This application is a clinical decision-support aid only.

Patient	Sample 02	Age	23
Assessment date	Removed	Reviewer	Removed
Eyes	OD; OS	Overall disposition	PASS WITH CAUTION

PASS WITH CAUTION — surgeon review required.

OD — PASS WITH CAUTION

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	0	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	0	
Manifest MRSE	0	
Total	0	
Topography category	NORMAL_SYMMETRIC	
Disposition	PASS	

NICE

Component	Input	Points
K2	45.4	2
central_pachymetry	533	1
B_Ele_Th	12	1
I_S	-0.25	1
Total		5
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 5

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 43.7 D < 48 D.
thinnest	NORMAL	Thinnest 531 µm >=500 µm.
elevation	NORMAL	F.Ele.Th 6 µm <=12 and B.Ele.Th 12 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 2/5; exceeded: anterior Km, front elevation at thinnest.
srax	NORMAL	Front-map SRAX 0.6°; signed I-S -0.25 D is negative (superior asymmetry), so SRAX does not create an inferior-steepening PS3 factor.
ppi_average	NORMAL	PPI Average 0.95 <= 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	0	0
Procedure disposition	PASS	Raw PS3 LASIK disposition: ALLOWED.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
NORMAL	0.05	9.3
Interpretation	Astigmatic disparity within validation thresholds: magnitude difference 0.05 D; axis difference 9.3°; no PS3 consequence.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	1.49	NORMAL / PASS
Df	0.97	NORMAL / < 1.60; information only
Db	0.68	NORMAL / < 1.60; information only
Dp	0.32	NORMAL / < 1.60; information only
Dt	0.21	NORMAL / < 1.60; information only
Da	0.38	NORMAL / < 1.60; information only
PPI Min	0.53	NORMAL / < 0.80; information only
PPI Avg	0.95	NORMAL / < 1.08; information only
PPI Max	1.19	NORMAL / < 1.40; information only
ARTmax	446	NORMAL / > 368 µm; information only

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_PTA_percent	31.191	
LASIK_RSB_um	365.375	
PRK_PTA_percent	Not documented	
PRK_RST_um	Not documented	
estimated_final_Kmean_D	40.2	
final_Kmean_model	MYOPIC_0.8_D_PER_D_ESTIMATE	
hard_stops	final_kmean: No; lasik_pta: No; lasik_rsb: No; preop_thickness: No; prk_pta: No; prk_rst: No; sphere_magnitude: No	
hyperopic_review_level	Not documented	
hyperopic_treatment_component_D	-2.75	
missing	Not documented	
scalar_final_Kmean_model_valid	Yes	
status	PASS	

Procedure planning

Planning item	Canonical result
OD — PASS WITH CAUTION — LASIK	Plan A — OZ 6.5 mm — TZ 9 mm — Flap 100 µm — Ablation 65.625 µm
MMC guidance	NOT_APPLICABLE
rejection_reasons	Not documented
Selected LASIK plan	Plan A — flap 100 µm; optical zone 6.5 mm; transition zone 9.0 mm
Plan-selection priority	Select the first safe plan only: Plan A → Plan B → Plan C
sequence	ablation_source: CERAI_MYOPIC_ESTIMATE; ablation_um: 65.625; definition: Plan A — flap 100 µm; optical zone 6.5 mm; transition zone 9.0 mm; flap_um: 100; optical_zone_mm: 6.5; plan: Plan A; rejection_reasons: Not documented; safe: Yes; status: PASS WITH CAUTION; transition_zone_mm: 9
ML7 alternative_hinge	Not documented
ML7 alternative_pta_percent	Not documented
ML7 alternative_rsb_um	Not documented
ML7 alternative_safety	NOT_APPLICABLE
ML7 applicable	Yes
ML7 assessment_gate	PASS WITH CAUTION
ML7 blade_recommendations	Plano/standard blade unless another rule or clinical factor applies
ML7 delta_k_d	3.2
ML7 hinge_location_alternative	Not documented
ML7 Preferred hinge location	Temporal
ML7 notes	Recommendation only; surgeon must verify anatomy, device setup, and the active ML7 manual before use.; If the selected ring leaks, does not hold vacuum, or is too large, the active ML7 reference directs selection of one smaller ring.
ML7 primary_hinge	Temporal hinge (default)
ML7 ring_tzone_clearance_mm	0
ML7 source	MED-LOGICS ML7 Rev. 22 active Turkish reference + CER-AI hinge amendment
ML7 status_independent	Yes
ML7 steep_meridian_axis_deg	Not documented
ML7 Vacuum pressure	550
ML7 Vacuum ring	9
ML7 warnings	Ablation/transition zone is not 0.4-0.5 mm smaller than the selected ring (active ML7 reference optimum).

Decision basis

Level	Canonical decision driver
CAUTION	detail: NICE total: 5; key: nice; status: CAUTION

Canonical Pentacam values and provenance

Canonical field	Value	Provenance
ARTmax	446	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
Final BAD-D	1.49	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
B.Ele.Th	12	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
CKI	1.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Da	0.38	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Db	0.68	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Df	0.97	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dp	0.32	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dt	0.21	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
F.Ele.Th	6	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
IHA	8.1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IHD	0.014	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
ISV	32	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IVA	0.15	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Signed I-S	-0.25	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
K1	42.2	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2	45.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2 axis	98.3	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
KI	1.02	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
KISA	1.59	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Kmax (Front)	45.7	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Km	43.7	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
PPI Avg	0.95	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PPI Max	1.19	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image

Canonical field	Value	Provenance
PPI Min	0.53	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PS3 BAD Axis (flat meridian, beside K1)	8.3	BELIN_AMBROSIO_CENTER_NUMERIC_BOX / De-identified source image
Pupil Center pachymetry	533	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
HWTW	11.5	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Thinnest pachymetry	531	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
SRAX	0.6	AXIAL_SAGITTAL_CURVATURE_FRONT_GEOMETRIC / De-identified source image
Astigmatism	3.2	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Displayed steep/astigmatic axis	98.3	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Topometric RMin	7.38	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image

Version provenance

Layer	Version
clinical_policy	CER-AI-2026-10-03-CAUTION-AGGREGATION-V12
deployment_sha	c15cccd4dfd9a4be162918141cebaec4b33ed77e
software	2.0.1
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07

OS — PASS WITH CAUTION

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	0	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	0	
Manifest MRSE	0	
Total	0	
Topography category	NORMAL_SYMMETRIC	
Disposition	PASS	

NICE

Component	Input	Points
K2	46	2
central_pachymetry	540	1
B_Ele_Th	11	1
I_S	-0.3	1
Total		5
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 5

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 44.1 D < 48 D.
thinnest	NORMAL	Thinnest 537 µm >=500 µm.
elevation	NORMAL	F.Ele.Th 4 µm <=12 and B.Ele.Th 11 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 2/5; exceeded: anterior Km, front elevation at thinnest.
srax	NORMAL	Front-map SRAX 5.8°; signed I-S -0.3 D is negative (superior asymmetry), so SRAX does not create an inferior-steepening PS3 factor.
ppi_average	NORMAL	PPI Average 0.9 <= 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	0	0
Procedure disposition	PASS	Raw PS3 LASIK disposition: ALLOWED.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
VALIDATION_REQUIRED	1.25	0.5
Interpretation	Astigmatic disparity (magnitude difference 1.25 D; axis difference 0.5°) requires measurement/refraction validation; it is not a PS3 risk factor and does not independently defer LASIK.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	1.54	NORMAL / PASS
Df	0.78	NORMAL / < 1.60; information only
Db	0.82	NORMAL / < 1.60; information only
Dp	-0	NORMAL / < 1.60; information only
Dt	0.02	NORMAL / < 1.60; information only
Da	0.54	NORMAL / < 1.60; information only
PPI Min	0.52	NORMAL / < 0.80; information only
PPI Avg	0.9	NORMAL / < 1.08; information only
PPI Max	1.25	NORMAL / < 1.40; information only
ARTmax	429	NORMAL / > 368 um; information only

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_PTA_percent	28.049	
LASIK_RSB_um	386.375	
PRK_PTA_percent	Not documented	
PRK_RST_um	Not documented	
estimated_final_Kmean_D	41.4	
final_Kmean_model	MYOPIC_0.8_D_PER_D_ESTIMATE	
hard_stops	final_kmean: No; lasik_pta: No; lasik_rsb: No; preop_thickness: No; prk_pta: No; prk_rst: No; sphere_magnitude: No	
hyperopic_review_level	Not documented	
hyperopic_treatment_component_D	-2.25	
missing	Not documented	
scalar_final_Kmean_mode_l_valid	Yes	
status	PASS	

Procedure planning

Planning item	Canonical result
OS — PASS WITH CAUTION — LASIK	Plan A — OZ 6.5 mm — TZ 9 mm — Flap 100 µm — Ablation 50.625 µm
MMC guidance	NOT_APPLICABLE
rejection_reasons	Not documented
Selected LASIK plan	Plan A — flap 100 µm; optical zone 6.5 mm; transition zone 9.0 mm
Plan-selection priority	Select the first safe plan only: Plan A → Plan B → Plan C
sequence	ablation_source: CERAI_MYOPIC_ESTIMATE; ablation_um: 50.625; definition: Plan A — flap 100 µm; optical zone 6.5 mm; transition zone 9.0 mm; flap_um: 100; optical_zone_mm: 6.5; plan: Plan A; rejection_reasons: Not documented; safe: Yes; status: PASS WITH CAUTION; transition_zone_mm: 9
ML7 alternative_hinge	Not documented
ML7 alternative_pta_percent	Not documented
ML7 alternative_rsb_um	Not documented
ML7 alternative_safety	NOT_APPLICABLE
ML7 applicable	Yes
ML7 assessment_gate	PASS WITH CAUTION
ML7 blade_recommendations	Plano/standard blade unless another rule or clinical factor applies
ML7 delta_k_d	3.6
ML7 hinge_location_alternative	Not documented
ML7 Preferred hinge location	Temporal
ML7 notes	Recommendation only; surgeon must verify anatomy, device setup, and the active ML7 manual before use.; If the selected ring leaks, does not hold vacuum, or is too large, the active ML7 reference directs selection of one smaller ring.
ML7 primary_hinge	Temporal hinge (default)
ML7 ring_tzone_clearance_mm	-0.5
ML7 source	MED-LOGICS ML7 Rev. 22 active Turkish reference + CER-AI hinge amendment
ML7 status_independent	Yes
ML7 steep_meridian_axis_deg	Not documented
ML7 Vacuum pressure	550
ML7 Vacuum ring	8.5
ML7 warnings	Ablation/transition zone is not 0.4-0.5 mm smaller than the selected ring (active ML7 reference optimum).

Decision basis

Level	Canonical decision driver
CAUTION	detail: NICE total: 5; key: nice; status: CAUTION

Canonical Pentacam values and provenance

Canonical field	Value	Provenance
ARTmax	429	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
Final BAD-D	1.54	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
B.Ele.Th	11	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
CKI	1.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Da	0.54	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Db	0.82	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Df	0.78	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dp	-0	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dt	0.02	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
F.Ele.Th	4	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
IHA	3.6	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IHD	0.013	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
ISV	35	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IVA	0.16	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Signed I-S	-0.3	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
K1	42.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2	46	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2 axis	87.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
KI	1.03	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
KISA	1.179	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Kmax (Front)	46.5	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Km	44.1	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
PPI Avg	0.9	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PPI Max	1.25	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image

Canonical field	Value	Provenance
PPI Min	0.52	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PS3 BAD Axis (flat meridian, beside K1)	177.5	BELIN_AMBROSIO_CENTER_NUMERIC_BOX / De-identified source image
Pupil Center pachymetry	540	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
HWTW	11.4	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Thinnest pachymetry	537	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
SRAX	5.8	AXIAL_SAGITTAL_CURVATURE_FRONT_GEOMETRIC / De-identified source image
Astigmatism	3.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Displayed steep/astigmatic axis	87.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Topometric RMin	7.26	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image

Version provenance

Layer	Version
clinical_policy	CER-AI-2026-10-03-CAUTION-AGGREGATION-V12
deployment_sha	c15cccd4dfd9a4be162918141cebaec4b33ed77e
software	2.0.1
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07